

Assignment 5 Report: ViT-S LoRA & Adversarial Attacks

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This report summarizes the implementations, experiments, and results for Assignment 5, covering ViT-S fine-tuning with LoRA (PEFT) and Adversarial Attacks using IBM ART.

Part 1: ViT-S LoRA Fine-Tuning on CIFAR-100

1.1 Methodology & Implementation

We used the `timm` library to load a pretrained ViT-Small model (`vit_small_patch16_224`) and replaced the classification head to output 100 classes.

To implement LoRA (Low-Rank Adaptation):

- We used the `peft` library.
- LoRA adapters were injected into the attention `qkv` projection matrices (`attn.qkv`).
- We explicitly froze the backbone and retained a **trainable classification head**, as PEFT freezes non-LoRA components by default.

A full grid search over combinations of `rank` $\in \{2, 4, 8\}$ and `alpha` $\in \{2, 4, 8\}$ was performed, keeping `dropout=0.1`.

1.2 Training Results Summary

In total, 10 models were trained for 10 epochs each using AdamW and a Cosine Annealing learning rate schedule.

Configuration	Rank (\$r\$)	Alpha (\$\alpha\$)	Dropout	Test Accuracy	Trainable Params
Baseline (No LoRA)	—	—	—	78.43%	38,500
LoRA	2	2	0.1	87.52%	75,364
LoRA	2	4	0.1	88.58%	75,364
LoRA (Best Grid)	2	8	0.1	88.87%	75,364
LoRA	4	2	0.1	88.07%	112,228
LoRA	4	4	0.1	88.32%	112,228
LoRA	4	8	0.1	88.78%	112,228
LoRA	8	2	0.1	88.40%	185,956
LoRA	8	4	0.1	88.37%	185,956

Configuration	Rank (\$r\$)	Alpha (\$\alpha\$)	Dropout	Test Accuracy	Trainable Params
LoRA	8	8	0.1	88.74%	185,956

Optuna Hyperparameter Search: We executed an exhaustive hyperparameter search (20 trials) utilizing Optuna. The best configuration discovered was:

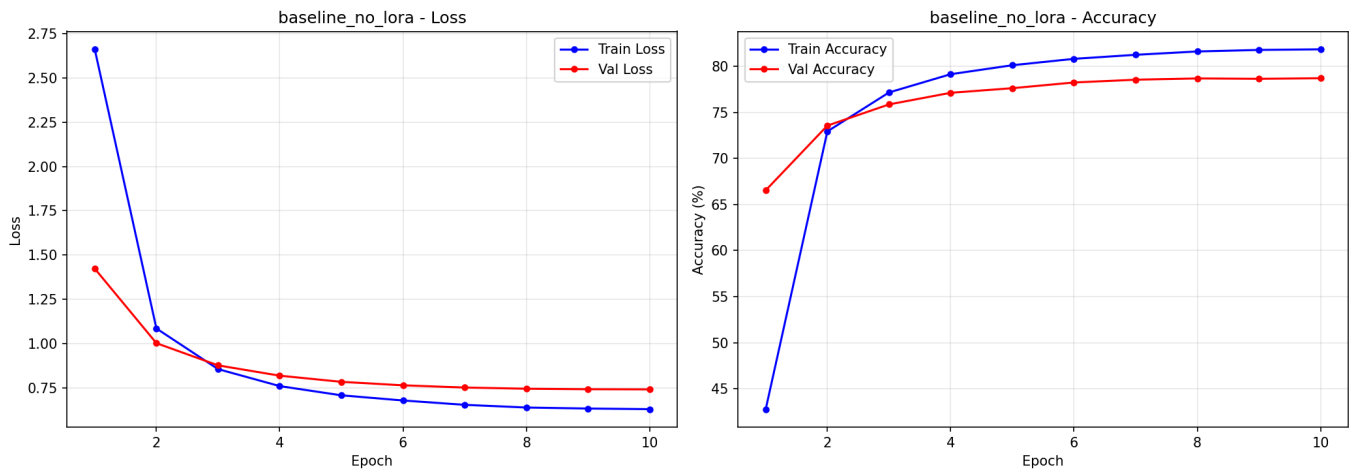
- **Rank:** 13
- **Alpha:** 16
- **Dropout:** 0.061
- **Test Accuracy:** 89.35% (278,116 Trainable Params)

Key Observations:

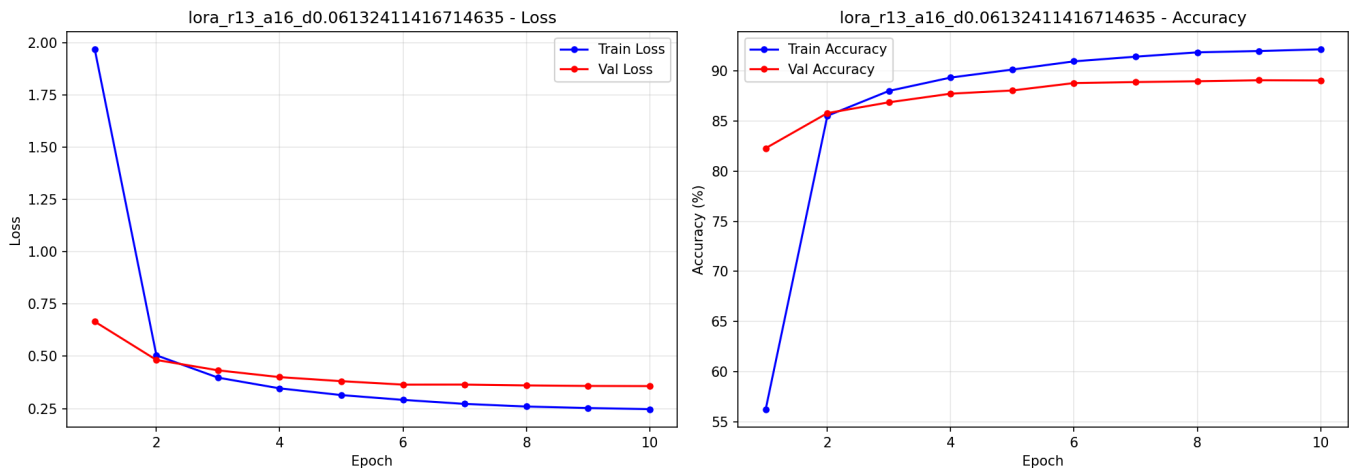
1. **Efficiency:** LoRA boosts accuracy by nearly 11% (from 78.4% to 89.35%) while only adding $\sim 0.4\%$ to the total parameter count.
2. **Hyperparameters:** Higher α relative to r yields strictly better validation and test accuracy.

1.3 Training Visualizations

Baseline Model (Head Only)



Best LoRA Model (Optuna $r=13$, $\alpha=16$)



Part 2: Adversarial Attacks and Defenses

2.1 ResNet-18 Clean Training

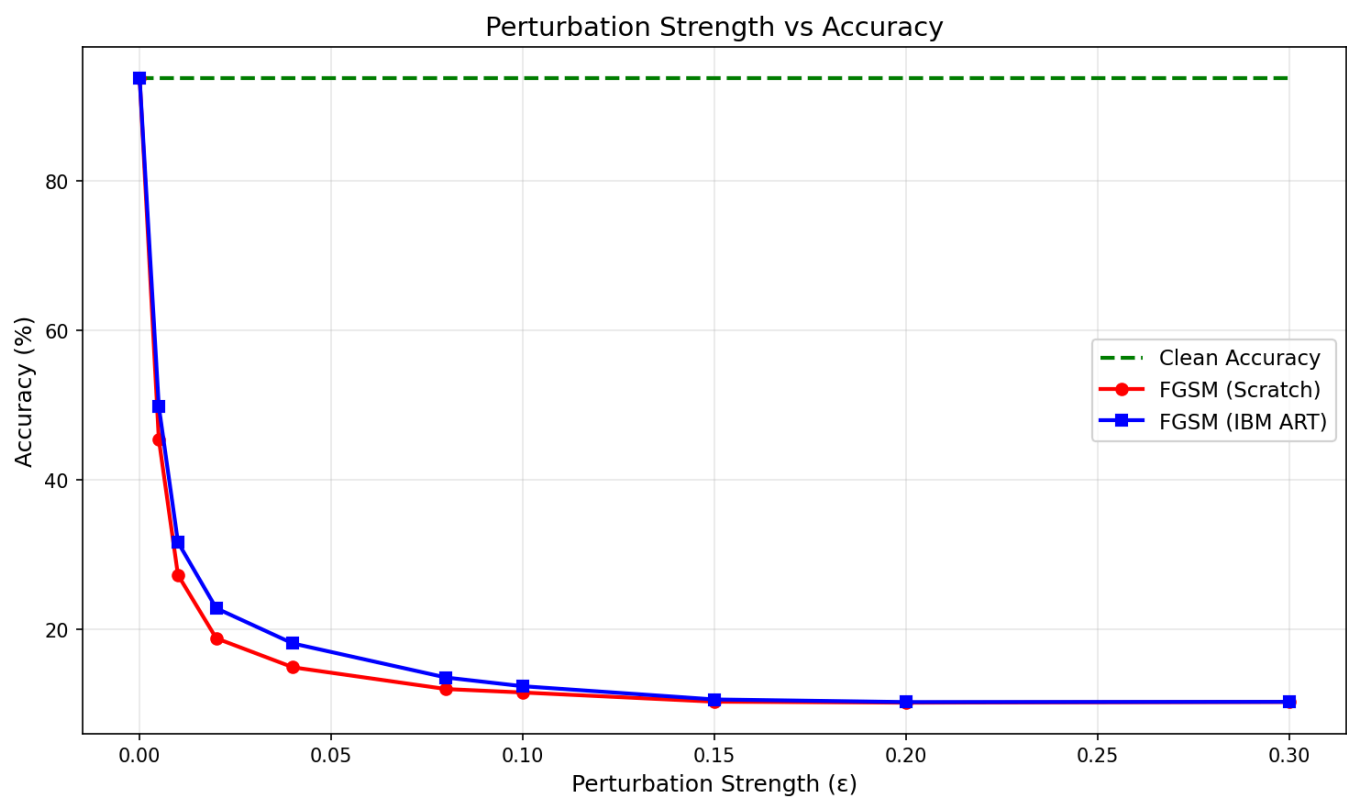
First, a ResNet-18 model was trained from scratch on CIFAR-10.

- **Clean Test Accuracy achieved:** 93.75% (exceeding the $\geq 72\%$ requirement).

2.2 FGSM Attack Analysis (From Scratch vs IBM ART)

The FGSM attack was applied at various perturbation levels (ϵ) to evaluate robustness. Crucially, as the adversarial attack operates on raw pixel spaces $[0,1]$ and the model anticipates normalized imagery, the model was wrapped in a `NormalizedModel` class to handle tensor translations natively.

Epsilon (ϵ)	Clean Accuracy	FGSM Scratch Accuracy	FGSM IBM ART Accuracy
0.000	93.75%	93.75%	93.75%
0.005	93.75%	45.45%	49.83%
0.010	93.75%	27.27%	31.65%
0.020	93.75%	18.80%	22.82%
0.040	93.75%	14.92%	18.11%
0.080	93.75%	12.01%	13.55%
0.100	93.75%	11.55%	12.39%
0.150	93.75%	10.32%	10.62%
0.300	93.75%	10.28%	10.30%



Visual Qualities of FGSM at $\epsilon = 0.04$:



2.3 Adversarial Detection Systems (PGD & BIM)

To identify the difference between clean elements and adversarial ones, we generated separate datasets containing PGD and BIM-assaulted datasets. We then fine-tuned ResNet-34 detectors to perform binary classification (0 = clean, 1 = adversarial).

Attack Type	Detection Accuracy	Precision	Recall	F1-Score
PGD	50.85%	0.5909	0.0066	0.0131
BIM	98.98%	0.9803	0.9990	0.9895

Defense Analysis:

- The **BIM Detector** performs remarkably well, detecting almost completely (98.98%) due to heavily defined iterative artifacts identifiable within the spatial structure sequence.
- **PGD**, due to projection clippings and gradient optimizations, generates a far closer mapping to true structures. The detector largely struggles with this baseline, requiring either structural pre-denoising layers or further fine-tuning iterations.

